Lucas Neumann

Senior Software Engineer — Pricing, Risk & AI Tooling

São Paulo, Brazil · Open to relocation / remote · neumann.m.lucas@gmail.com · linkedin.com/in/lucas-mneumann · github.com/neumann-mlucas · neumann-mlucas.github.io/cv

Summary

Senior backend engineer (Python, Go) building data pipelines, cloud infrastructure and AI-agent tooling for financial systems. Designed MCP-based tooling that gives LLM agents safe, read-only access to production data, and owned real-time market-data and pricing services where downtime is not an option.

Experience

BWGI 2025–present

Senior Software Engineer

- Moved portfolio sourcing to vendor PMS platforms (Arcesium) and built an automated tandem reconciliation of legacy vs. new pricing, classifying breaks with two-level tolerances; in production since 2026.
- Designed a Go toolkit of 15+ read-only CLIs and **MCP** servers that let LLM agents (Claude Code, Codex) safely query databases, pipelines, market data and official risk metrics: read-only SQL gates, replica routing, audit log with kill switch, secrets kept out of agent context.
- Wrote 20+ agent skills backed by a docs-as-tests harness, shipped versioned toolkit releases to the risk team and added AI code review to CI.
- Migrated risk and pricing workflows from Airflow 2 to Astronomer (Airflow 3 on Kubernetes) and upgraded 18 shared packages to Pandas 2, NumPy 2 and SQLAlchemy 2.
- Main on-call for pricing and risk; about 790 merged PRs across 32 repositories in 11 months.

Bank of America 2023–2025

Software Engineer, AVP — FICC Emerging Markets

Joined via BRQ as consultant in 2023; hired directly as Assistant Vice President in 2024.

- Built and ran real-time curve infrastructure for the EM derivatives and FX trading desk: live market data and trader curve marks feeding the sell-side pricing system, plus intraday and end-of-day curve publication to internal databases.
- Owned the real-time **PTAX** integration with Banco Central do Brasil — a high-availability service with near-zero fault tolerance, where any failure carries direct reputational risk.
- Developed in one of the industry's largest Python monorepos (15+ years, C++ performance layer): event-driven market-data processing on proprietary NoSQL and messaging systems.

Apex Capital 2021–2022

Software Developer

Hedge fund

- Migrated operational routines from Windows Task Scheduler to Apache Airflow, improving reliability and visibility of daily fund processes.
- Integrated financial-service APIs and built dashboards, data visualisations and scrapers for the investment team.

Skills

Engineering
Python, Go, SQL (PostgreSQL, Snowflake), Airflow/ Astronomer, Docker, Kubernetes, AWS, Datadog, MCP, Linux, Git

Financial systems
Bloomberg MARS & market data, Arcesium, MSCI Barra, S&P Global, Qlik

Domain
Rates, credit, FX, fixed income; derivatives pricing; VaR, expected shortfall, Greeks; reconciliation

Languages
Portuguese (native), English (fluent)

Education

UNICAMP — University of Campinas
MSc, Physical Chemistry
2018–2020
Computational simulation of proteins; daily work with HPC clusters, Python, Julia, C/C++ and Linux.

Projects

git-archaeologist (Rust) — Fast CLI extracting time series from git history (LOC, cohort survival, coupling, hotspots) as TSV, CSV, JSON or Parquet.

ragwatcher (Python) — Incremental local RAG over a directory, served as an MCP tool and CLI.